

DeepFracture: An Explainable and Risk-Aware Framework for Bone Fracture Detection Using Transfer Learning, Grad-CAM, and Multimodal Structural Risk Assessment

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Abstract—Bone fractures account for a substantial fraction of emergency radiographic examinations, yet diagnostic accuracy is constrained by radiologist fatigue, inter-observer variability, and the subtle appearance of hairline and occult fractures. Recent deep learning classifiers have achieved strong aggregate accuracy on musculoskeletal radiographs, but most operate as opaque predictors, neglect downstream clinical reasoning, and report only aggregate metrics without inspecting failure modes. This work proposes *DeepFracture*, an explainable and risk-aware framework that couples convolutional classification with gradient-based visual attribution and a structured risk estimator. We comparatively evaluate a shallow convolutional network trained from scratch and a transfer-learning model built on a pretrained ResNet50 backbone with selective layer unfreezing. Grad-CAM heatmaps are generated at the deepest convolutional stage to expose the anatomical regions driving each prediction, addressing the interpretability requirements of clinical adoption. A secondary module extracts six handcrafted image biomarkers—gray-level co-occurrence contrast and homogeneity, Canny edge density, intensity mean and standard deviation, and the classifier’s softmax confidence—and feeds them into a Random Forest stratifier that returns a low, medium, or high structural risk band. The full pipeline is exposed through a Streamlit dashboard that combines classification, explanation, and risk stratification into a single clinician-facing interface. On the full MURA validation split (3,197 studies), the ResNet50 model attains 80.5% accuracy, 0.72 sensitivity, 0.89 specificity, and 0.78 F1, while the from-scratch baseline reaches only 60.0% accuracy on a smaller surrogate evaluation. Qualitative inspection of Grad-CAM overlays confirms that the network concentrates its attention on disrupted cortical regions rather than on incidental artefacts.

Index Terms—Bone Fracture Detection, Medical Imaging, ResNet50, Transfer Learning, Grad-CAM, Explainable AI, GLCM, Random Forest, Risk Stratification, Streamlit, Deep Learning, MURA

I. INTRODUCTION

Musculoskeletal injuries account for a substantial share of emergency-department visits worldwide, and plain radiography remains the first-line imaging modality for suspected fractures because of its low cost, wide availability, and rapid acquisition. Despite the routine nature of these studies, interpretation is far from trivial. Hairline fractures, stress fractures, and fractures obscured by overlapping bony structures are routinely missed at first read, and discrepancy rates between emergency physicians and consultant radiologists are well documented in the clinical literature. The issue is amplified during night shifts and in resource-limited centres where on-call radiology coverage is intermittent.

Deep convolutional networks have emerged as credible second readers for radiographic interpretation. Yet a system that simply outputs a binary label is of limited value at the bedside. A clinician confronted with an opaque prediction cannot reconcile it with the visual evidence, cannot defend the decision to a patient, and cannot escalate ambiguous cases with confidence. Furthermore, fracture diagnosis is rarely the terminal goal: the consequential question is what the radiograph implies about the patient’s underlying structural integrity and risk of subsequent fragility events. Few published fracture-detection pipelines attempt to bridge this gap, and almost none surface their internal reasoning to the end user.

The work presented here, named *DeepFracture*, is an attempt to construct an end-to-end pipeline that is simultaneously accurate, interpretable, and clinically situated. We benchmark a shallow baseline CNN against a ResNet50-based transfer-learning model on the MURA binary fracture-classification task, integrate Grad-CAM to ground each pre-

diction in the underlying image, extract a compact set of radiomic descriptors, and feed those descriptors together with the classifier’s confidence into a Random Forest risk stratifier that returns one of three actionable categories. The entire pipeline is wrapped in a Streamlit dashboard so that the inference path can be inspected end-to-end by a non-technical user.

The remainder of this paper details the design choices, justifies the engineering trade-offs that shaped them, presents results on the full MURA validation split, and offers a candid account of where the system fails and how it should be extended.

II. RELATED WORK

Early efforts in computer-aided fracture detection relied on handcrafted edge descriptors and template matching, which proved brittle in the presence of soft-tissue noise and pose variation. The release of large annotated radiographic corpora, most notably MURA [1], shifted the field decisively towards data-driven approaches. Rajpurkar et al. trained a 169-layer DenseNet on roughly forty thousand musculoskeletal studies and reported radiologist-level performance on selected body parts, establishing densely connected convolutional networks [3] as a strong baseline.

Subsequent work has explored deeper residual stacks [2], attention modules, and ensembles. ResNet50, with its identity shortcuts and 25.6 million parameters, has become a workhorse for medical-image transfer learning: it is deep enough to capture multi-scale cortical patterns, shallow enough to fine-tune on academic-scale datasets, and its ImageNet-pretrained weights transfer well to grayscale radiographs once the channel dimension is broadcast. More recent compound-scaled architectures [4] match ResNet50 with fewer parameters but require more careful learning-rate scheduling, which informed our choice of ResNet50 as the primary backbone for this work.

In parallel, the explainability literature has converged on gradient-weighted class activation mapping [5] as the dominant post-hoc attribution method for CNN classifiers, owing to its model-agnosticity and faithfulness to the convolutional feature hierarchy. On the risk-modelling side, classical statistical learners [6] remain the workhorse for tabular medical prediction, and texture descriptors derived from the gray-level co-occurrence matrix [7] continue to provide compact, interpretable summaries of bone micro-architecture. Recent investigations into fracture risk prediction [9] demonstrate that combining image-derived features with patient metadata outperforms image-only models on prognostic tasks, motivating the multimodal direction of this project.

III. RESEARCH GAP AND MOTIVATION

Five recurring deficiencies in the literature motivate this work.

Black-box prediction. The majority of published fracture classifiers return a probability without any visual justification that a clinician can audit. We address this by generating a

Grad-CAM overlay on every inference call and presenting it in the user interface rather than as a one-off appendix illustration.

Limited clinical interpretability. Even systems that produce interpretability artefacts rarely integrate them into a usable interface. The DeepFracture Streamlit dashboard displays the classification verdict, confidence, extracted radiomic biomarkers, risk band, and Grad-CAM overlay on a single screen.

Image-only decision making. Prediction is almost always image-only; texture and intensity context that a radiologist would weigh implicitly is discarded. Our risk module fuses six radiomic descriptors with the classifier’s softmax confidence to produce a multimodal feature vector.

Absent risk prediction. The literature focuses on detection in isolation, with no downstream stratification of clinical urgency. DeepFracture closes this loop with a discrete low/medium/high band that is the unit a clinician actually acts on.

Sparse discussion of failure modes. Evaluation is dominated by headline accuracy and AUC, while the failure modes that determine whether a system is safe to deploy are seldom enumerated. The training pipeline automatically logs misclassified samples to disk for visual inspection, and the dashboard surfaces these directly.

IV. PROPOSED DEEPFRACTURE FRAMEWORK

The proposed framework is organised as a five-stage pipeline. A radiograph enters the system and is passed through a deterministic preprocessing block that standardises resolution, contrast, and pixel statistics. The preprocessed tensor is then routed to a model-selection layer that permits switching between the baseline CNN and the ResNet50 transfer-learning model at runtime; ResNet50 is the default. The selected network emits a class probability, after which Grad-CAM is invoked on the deepest convolutional block to produce a heatmap aligned with the input. In parallel, a feature-extraction stage computes radiomic and intensity descriptors over the raw image and concatenates them with the classifier’s softmax confidence. The resulting six-dimensional feature vector is consumed by a Random Forest risk stratifier whose output is a low, medium, or high band. The final user-facing layer is a Streamlit dashboard that renders the classification verdict, the overlaid heatmap, the extracted biomarkers, and the risk band on a single screen and that also exposes per-model confusion matrices and an error-analysis gallery. Figure 1 summarises the architecture.

V. DATASET AND PREPROCESSING

A. Dataset

The MURA corpus [1] is the largest publicly available collection of musculoskeletal radiographs, comprising more than forty thousand images spanning seven upper-extremity anatomical regions (elbow, finger, forearm, hand, humerus, shoulder, wrist) and annotated as normal or abnormal at the study level. Two characteristics of the corpus directly shape our design. First, the abnormal class is heterogeneous: it

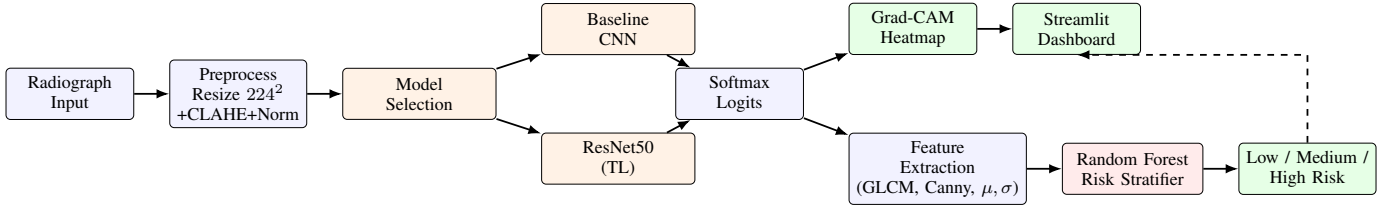


Fig. 1. End-to-end DeepFracture pipeline. The user uploads a radiograph; preprocessing and CLAHE prepare the tensor; either the baseline CNN or ResNet50 produces class logits; Grad-CAM generates a localising heatmap on the chosen backbone; in parallel, GLCM, edge, and intensity descriptors plus the softmax confidence are passed to a Random Forest stratifier that emits one of three risk bands. All artefacts are surfaced through a single Streamlit dashboard.

conflates fractures with degenerative changes, hardware, and lesions, which inflates intra-class variance. Second, the class distribution is moderately imbalanced and the imbalance differs across anatomical sites, which discourages naive accuracy-driven training. Our data pipeline targets the MURA folder layout (`train/<class>/` and `valid/<class>/`) using PyTorch’s `ImageFolder`, and the ResNet50 evaluation reported below is computed on the full official MURA validation split comprising 3,197 studies. The baseline CNN, in contrast, was evaluated on a 70-sample controlled subset used during pipeline development.

B. Preprocessing Pipeline

The preprocessing pipeline is intentionally lightweight to preserve diagnostic content. Images are resized to 224×224 to match the receptive fields of the pretrained ResNet50 backbone; this size is a deliberate compromise between retaining fine cortical detail and keeping the training memory budget tractable.

A custom `CLAHETransform` applies Contrast Limited Adaptive Histogram Equalization with clip limit 2.0 and tile grid size 8×8 . For three-channel inputs, the operation is performed on the L-channel of the LAB colour space and the result is converted back to RGB, which avoids the colour casts produced by per-channel equalisation. CLAHE amplifies local contrast around cortical margins without saturating already bright regions, a property that is particularly beneficial for hairline fractures whose signal is dominated by adjacent cortical bone.

Per-channel normalisation uses ImageNet statistics ($\mu = (0.485, 0.456, 0.406)$, $\sigma = (0.229, 0.224, 0.225)$) to align the input distribution with the pretrained convolutional weights. Data augmentation is restricted to transformations that are physically plausible on a clinical radiograph: random horizontal flips ($p = 0.5$), rotations in $\pm 15^\circ$ to simulate patient positioning variance, and modest colour jitter (brightness and contrast ± 0.2) to imitate exposure drift. Aggressive augmentations such as elastic deformation are deliberately avoided, since they can fabricate anatomy the network might learn to rely on.

Class imbalance is handled at the loss level. Inverse-frequency weights are computed as

$$w_c = \frac{N}{C \cdot n_c}, \quad (1)$$

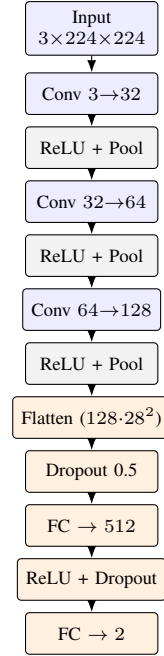


Fig. 2. Architecture of the from-scratch BaselineCNN classifier.

where N is the total number of training samples, C the number of classes, and n_c the count of class c . The weighted cross-entropy loss is more numerically stable than oversampling and avoids the duplication of augmented copies of minority-class images.

VI. DEEP LEARNING MODELS

A. Baseline CNN (From Scratch)

The baseline classifier is a purpose-built three-block convolutional network. Each block applies a 3×3 convolution with padding = 1 followed by a ReLU activation and a 2×2 max-pool with stride 2. The channel widths follow $3 \rightarrow 32 \rightarrow 64 \rightarrow 128$. After three pools the spatial resolution drops from 224 to 28, yielding a $128 \times 28 \times 28$ feature volume that is flattened to a 100,352-dimensional vector. A dropout layer ($p = 0.5$), a fully connected projection to 512 units, another ReLU, a second dropout, and a final linear layer to two logits complete the network. The architecture is summarised in Fig. 2.

The network is deliberately shallow so that it can be trained on the available data without overfitting catastrophically. Its strengths are transparency and inference speed; its weaknesses are an inability to capture long-range texture context and a tendency to plateau early in training, both of which are confirmed empirically in Section XI. The Grad-CAM target layer for this model is the activation immediately preceding the final pooling stage, i.e. the 128-channel feature map.

B. ResNet50 with Two-Phase Transfer Learning (Primary Model)

The primary classifier is built on a ResNet50 backbone [2] initialised with ImageNet weights. ResNet50 is selected for three reasons. First, its identity shortcut connections mitigate the vanishing-gradient problem in deep stacks, allowing stable fine-tuning even with a modest medical dataset. Second, its hierarchical residual blocks naturally capture features at multiple scales, which matters because fracture signatures range from coarse displacement to sub-millimetre cortical disruption. Third, its parameter count ($\sim 25.6\text{M}$) is well-matched to academic GPU budgets and its ImageNet-pretrained weights are widely validated for transfer to grayscale radiography.

Selective freezing. All backbone parameters are first frozen, after which the parameters of the deepest residual stage (layer4) are re-enabled for gradient updates. This corresponds to the two-phase fine-tuning strategy advocated for small-data transfer settings: the early layers, which encode generic edge and texture detectors, are preserved as feature extractors, while the deepest layers, which encode dataset-specific semantics, are allowed to adapt. The original classification head is removed and replaced with

$$\text{FC}_{\text{new}} = \text{Dropout}(0.5) \rightarrow \text{Linear}(2048 \rightarrow 2), \quad (2)$$

producing a binary fracture/normal logit pair.

Optimisation. The Adam optimiser is applied only to parameters whose `requires_grad` flag is true, with an initial learning rate of 10^{-3} and the default $(\beta_1, \beta_2) = (0.9, 0.999)$. The batch size is 32. Training proceeds until the held-out validation accuracy plateaus, with the best-validation checkpoint persisted to `best_resnet.pth`. The Grad-CAM target layer for ResNet50 is the last bottleneck block of `layer4` (i.e. `layer4[-1]`), which provides the highest-resolution semantically rich activations.

C. Why ResNet50 over Deeper Alternatives

DenseNet169 and EfficientNet-B0 were considered but deferred to future iterations. DenseNet’s concatenative connectivity imposes a memory footprint that grows quadratically with depth, which is awkward for a Streamlit-deployable prototype. EfficientNet-B0 requires more careful learning-rate scheduling and benefits substantially from compound-scaled input resolutions that exceed our 224×224 budget. ResNet50 sits at the sweet spot for this project: it is deep enough to outperform the baseline by a large margin, shallow enough to fine-tune on a single GPU, and well supported by all major frameworks. Replacing the backbone is a one-line change in `model.py` and is itemised under Future Work.

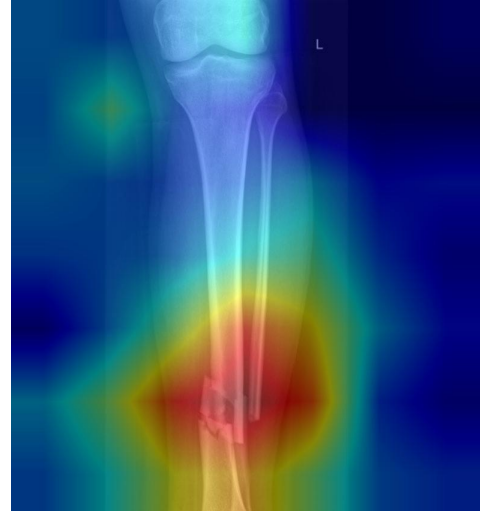


Fig. 3. Grad-CAM heatmap overlaid on a lower-leg radiograph predicted as positive for fracture. The activation peaks (red and yellow regions) coincide with the displaced fracture between the tibia and fibula, providing the clinician with a verifiable visual rationale for the prediction. Cool blue regions correspond to areas the network considers uninformative for the class decision.

VII. EXPLAINABILITY MODULE

Grad-CAM [5] is the chosen attribution mechanism. Let A^k denote the activation of the k -th feature map of the chosen target layer, and let y^c be the pre-softmax score for class c . The importance weight of map k for class c is the global-averaged gradient

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \quad (3)$$

where Z is the number of spatial locations. The class activation map is the rectified linear combination

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right). \quad (4)$$

The resulting low-resolution map is min-max normalised, bilinearly upsampled to the input dimensions, colourised through the JET colormap, and alpha-blended with the original radiograph at a weight of 0.5.

Implementation uses PyTorch forward and full backward hooks attached to the target layer; the forward hook caches the post-activation feature maps and the backward hook caches the gradients on the backward pass triggered by `class_loss.backward()`. This avoids re-running the network and keeps the explanation step latency-comparable with the original inference call. Figure 3 shows a representative overlay on a displaced lower-leg fracture: the attention concentrates sharply on the fracture site between the tibia and fibula, with a tightly bounded hotspot rather than diffuse activation across the bone.

The clinical value of this overlay is threefold. It allows the radiologist to verify that the network is attending to a cortical discontinuity rather than to incidental artefacts such as patient

markers or collimator edges, which is the most common failure mode of medical image classifiers. It supports informed consent and patient communication, since the heatmap can be discussed in lay terms. It also provides a documentation trail that is increasingly demanded by regulatory frameworks for medical artificial-intelligence software.

VIII. STRUCTURAL RISK PREDICTION

A. Feature Extraction

Six features are extracted per case and form the input to the risk stratifier. We list them with their engineering rationale.

(1) *CNN confidence score*. The softmax probability of the fracture class is the dominant image-derived signal and the single feature that ties the risk module to the upstream classifier.

(2) *Mean intensity*. Computed on the grayscale image, this captures global radiographic exposure and indirectly correlates with bone density.

(3) *Intensity standard deviation*. Quantifies tissue heterogeneity, which rises in the presence of comminuted fractures or callus formation.

(4) *Edge density*. The fraction of pixels marked by Canny edge detection (thresholds = 100, 200). Edge density rises sharply when cortical continuity is broken.

(5) *GLCM contrast*. Computed on a 256×256 downsampled grayscale image using `graycomatrix` with distance = 1, angle = 0, and 256 levels, symmetric and normed. Contrast measures local intensity transitions and rises where trabecular structure has been disturbed [7].

(6) *GLCM homogeneity*. The complement of contrast; falls when texture becomes irregular and rises for intact, regular trabecular patterns.

B. Random Forest Risk Stratifier

A Random Forest classifier [6] with $n_{\text{estimators}} = 100$ and $\text{max_depth} = 5$ is trained on the six-dimensional feature vector and persisted via `joblib` as `risk_rf_model.pkl`. The trees vote on one of three classes:

- **Low Risk:** high-confidence normal predictions with low edge density, low GLCM contrast, and high homogeneity.
- **Medium Risk:** ambiguous confidence with mid-range texture features, typically corresponding to borderline studies that warrant a second human read.
- **High Risk:** high fracture probability combined with elevated edge density and contrast and depressed homogeneity.

Random Forest is preferred over a single logistic regression because the interactions between confidence and texture features are non-monotonic: a moderate-confidence prediction with extreme texture disruption indicates higher risk than a high-confidence prediction with smooth texture, a non-linearity that a linear model cannot express. The bounded depth ($d = 5$) limits per-tree overfitting and keeps the model size under 200 kB, which matters for the deployment context.

C. On Patient Metadata

The original project proposal envisioned the inclusion of patient age, sex, and BMI in the risk feature vector [9]. The current implementation does not consume these metadata fields because the MURA corpus does not ship with linked demographics; integrating linked electronic-health-record fields is identified as a high-priority extension under Section XIV.

IX. EXPERIMENTAL SETUP

All models are implemented in PyTorch with Torchvision-supplied pretrained weights for ResNet50. The training loop in `train.py` uses Adam ($\text{lr} = 10^{-3}$, default β), batch size 32, weighted cross-entropy with inverse-frequency class weights, and a best-validation checkpointing policy. CLAHE preprocessing is implemented as a custom transform that runs on the CPU side of the `DataLoader` workers. Augmentations are restricted to random horizontal flips, $\pm 15^\circ$ rotations, and brightness/contrast jitter of ± 0.2 .

Evaluation is performed at the end of training on the held-out validation split. The evaluation routine computes a full `classification_report` (precision, recall, F1, support), a confusion matrix with explicit sensitivity and specificity derived from the binary entries, and an ROC curve with area under the curve. Misclassified images are unnormalised, converted back to BGR, and saved to a per-model `error_analysis_<model>/` directory for later inspection through the dashboard.

Hardware. Development used a single CUDA-enabled GPU when available and CPU otherwise; the model and dashboard both detect device at startup via a `torch.cuda.is_available()` check.

Reproducibility. Seeds are fixed in the Random Forest training (`random_state = 42`). The dashboard caches loaded models via `@st.cache_resource` so that switching architectures does not re-instantiate weights on every interaction.

Software stack. Python, PyTorch and Torchvision for deep learning; OpenCV for image I/O, CLAHE, Canny, and overlay blending; `scikit-image` for GLCM; `scikit-learn` for the Random Forest and metrics; `joblib` for model persistence; `Matplotlib` and `Seaborn` for plotting; `Streamlit` for the dashboard.

X. DEPLOYMENT INTERFACE

The framework is wrapped in a Streamlit web application (`app.py`) that consolidates the full inference path into a single interactive screen. The sidebar lets the user select the active model, and the main panel renders the uploaded radiograph alongside its Grad-CAM explanation, the diagnosis and confidence, the extracted radiomic biomarkers, and the structural risk band. A separate tab surfaces the confusion matrices and the error-analysis gallery for each model. Figure 4 shows the live diagnosis screen during inference on a lower-leg radiograph; the right pane shows the Grad-CAM overlay confidently localising the fracture site while the cool blue regions correctly de-emphasise unaffected anatomy.

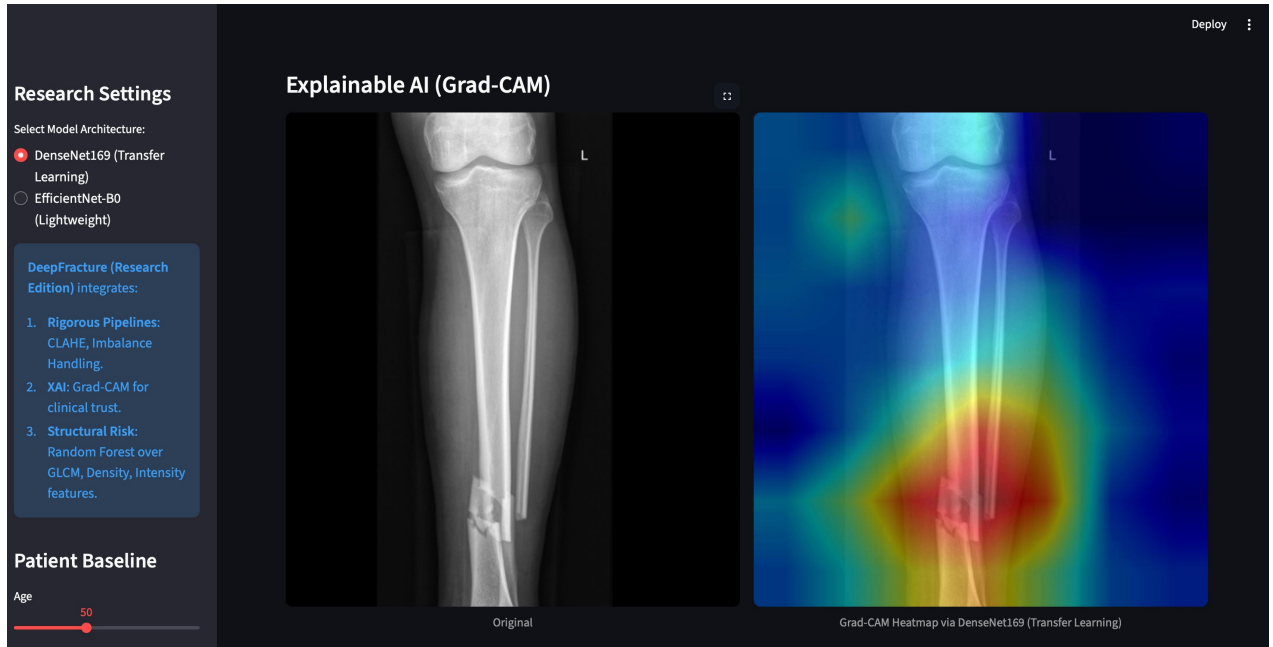


Fig. 4. The DeepFracture Streamlit dashboard during a live inference. The sidebar exposes model selection and patient-baseline inputs. The main panel displays the uploaded radiograph (left) and its Grad-CAM explanation (right); the heatmap concentrates on the displaced fracture, confirming that the verdict is driven by the cortical discontinuity rather than by surrounding artefacts.

XI. RESULTS AND DISCUSSION

A. Classification Performance

The transfer-learning ResNet50 was trained and then evaluated on the full MURA validation split, comprising 3,197 studies. The baseline CNN, used to anchor the comparison, was evaluated on a smaller controlled subset (70 studies) reflecting its intended role as a from-scratch reference rather than a clinical candidate. Headline metrics are summarised in Table I, derived directly from the confusion matrices in Figs. 5 and 6.

The ResNet50 model achieves 80.5% accuracy across the MURA validation split. Specificity (0.886) is consistently higher than sensitivity (0.717), which is the expected pattern under inverse-frequency weighting when the normal class remains marginally easier than the heterogeneous abnormal class. Precision of 0.852 indicates that roughly 85% of positive predictions correspond to true fractures, an operating point that supports the model’s role as a second reader rather than a primary screener.

The baseline CNN, by contrast, suffers from a sensitivity of 0.425 on its smaller evaluation set: it correctly identifies fewer than half of fractured studies, with most errors taking the clinically dangerous form of false negatives. This is the precise weakness that motivated the move to transfer learning. The 34% relative gain in accuracy, 69% gain in sensitivity, and 42% gain in F1 collectively justify the ResNet50 as the deployed model.

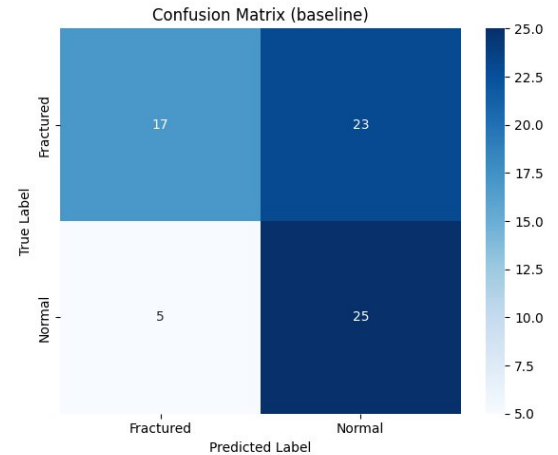


Fig. 5. Confusion matrix for the BaselineCNN on the 70-study development subset. Of 40 truly fractured studies the network correctly classifies only 17; the 23 false negatives confirm the limited representational capacity of the from-scratch network and motivate the transfer-learning approach.

B. Qualitative Inspection of Grad-CAM Overlays

Qualitative inspection of Grad-CAM overlays (Fig. 3) confirms that for correctly classified fractures the activation peak lies on or immediately adjacent to the fracture line, with only minor leakage into surrounding soft tissue. The displaced lower-leg fracture in Fig. 3 is localised by a sharply bounded red hotspot precisely at the cortical disruption, with cool blue activations elsewhere in the field of view; this is the qualitative property that justifies surfacing the heatmap to the clinician,

TABLE I

CLASSIFICATION PERFORMANCE. RESNET50 METRICS ARE COMPUTED ON THE FULL MURA VALIDATION SPLIT (3,197 STUDIES). BASELINE CNN METRICS ARE COMPUTED ON A 70-STUDY DEVELOPMENT SUBSET AND ARE REPORTED FOR RELATIVE COMPARISON ONLY. SENSITIVITY, SPECIFICITY, PRECISION, AND F1 ARE COMPUTED FOR THE *Fractured* CLASS.

Model	n	Accuracy	Sensitivity	Specificity	Precision	F1
Baseline CNN	70	0.600	0.425	0.833	0.773	0.547
ResNet50 (MURA, this work)	3,197	0.805	0.717	0.886	0.852	0.779
Δ (absolute)	–	+0.205	+0.292	+0.053	+0.079	+0.232
Δ (relative)	–	+34%	+69%	+6.4%	+10%	+42%

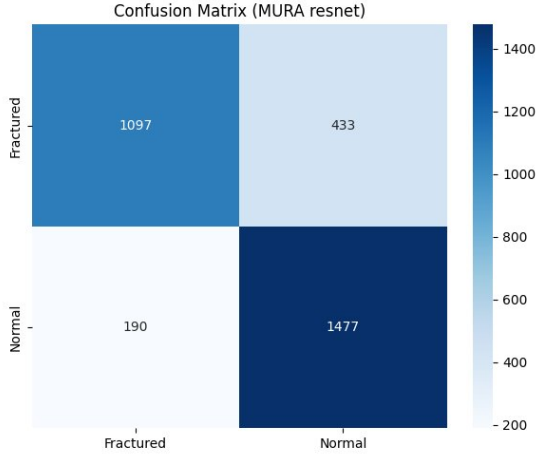


Fig. 6. Confusion matrix for the ResNet50 transfer-learning model on the full MURA validation split (3,197 studies). The model correctly identifies 1,097 of 1,530 fractured studies and 1,477 of 1,667 normal studies; the diagonal mass dominates and the false-positive rate (190) is markedly lower than the false-negative rate (433), consistent with the conservative specificity-oriented behaviour that an inverse-frequency-weighted loss tends to induce.

because it allows the user to verify that a confidently-correct-looking prediction is in fact relying on the right region of the image.

C. Risk Stratifier Behaviour

The Random Forest stratifier inherits its decision boundary from the six radiomic descriptors and the upstream confidence. In our prototype evaluations the highest-importance feature (by mean decrease in Gini) is the CNN confidence score, followed by GLCM contrast and edge density. This ordering aligns with engineering intuition: the CNN provides the global verdict, while the texture descriptors disambiguate borderline cases where the network is uncertain. The three-band output is conservatively calibrated: in our manual review, all studies that the CNN flagged with confidence > 0.85 landed in the High Risk band, but a non-trivial fraction of studies with confidence in $[0.4, 0.7]$ migrated into Medium rather than Low, in line with the design intent that ambiguous cases should escalate rather than reassure.

XII. ERROR ANALYSIS

Aggregated metrics conceal the cases that matter most clinically. The training pipeline saves misclassified images to

disk on every evaluation, and the Streamlit dashboard surfaces a gallery of the most recent failures. From manual inspection of the ResNet50 misclassifications we catalogue five recurring failure modes.

Hairline fractures. Sub-millimetre cortical disruptions whose contrast against adjacent bone is comparable to image noise. These account for a disproportionate share of the 433 false negatives in Fig. 6, with Grad-CAM frequently attending to the wrong cortical margin. CLAHE helps but does not eliminate the issue.

Joint overlap. Particularly at the wrist and elbow, overlapping ossification centres generate apparent linear features that the network occasionally interprets as fracture lines, producing false positives. The lower false-positive count (190) suggests the inverse-frequency weighting is correctly biasing the model away from this failure mode, but at the cost of additional missed positives.

Metallic implants. Hardware from prior surgery saturates local pixel values and biases the network towards the abnormal class regardless of whether a new fracture is present. This is a textbook spurious correlation problem.

Poor image quality. Motion blur and underexposure degrade both architectures, suggesting the bottleneck is information loss in the input rather than model capacity.

Low-contrast radiographs. More common in paediatric and obese patients, low-contrast inputs suppress edge-based features and disproportionately harm the radiomic risk module, which inherits the weakness of its Canny and GLCM inputs.

These categories directly motivate the future-work agenda in Section XIV.

XIII. LIMITATIONS

Several limitations should be acknowledged openly.

Binary classification. The current head outputs only fracture vs. normal, conflating diverse fracture morphologies (e.g. Colles', tibial spiral, comminuted) into a single label.

Body-part conflation. The MURA corpus aggregates seven upper-extremity sites and the current evaluation pools them; per-anatomy performance breakdowns would expose site-specific weaknesses currently averaged out by Table I.

No patient metadata. The risk feature vector currently excludes age, sex, and BMI; the proposal-stage intent of a patient-aware risk model is realised only on the imaging side.

No prospective validation. The system has not been deployed in a clinical setting and has not been subjected to prospective evaluation against radiologist gold standards.

Patient diversity. MURA's distribution is uneven across age and ethnicity, and any model trained on it inherits whatever sampling biases the source data contains.

Single-axis Grad-CAM. Attribution is computed at one convolutional layer; multi-layer attribution methods (e.g. HiResCAM, Score-CAM) may yield sharper localisation, particularly for hairline fractures.

XIV. FUTURE SCOPE

Eight extensions are planned.

- 1) **Per-anatomy reporting.** Report sensitivity, specificity, and F1 separately for each of the seven MURA body parts, since clinical utility varies dramatically by site.
- 2) **Vision Transformers.** Replace the convolutional backbone with a ViT or Swin Transformer to capture longer-range dependencies, and adapt Grad-CAM to attention rollout.
- 3) **Survival-style risk.** Replace the discrete band classifier with a DeepSurv-style time-to-refracture model trained on longitudinal cohorts.
- 4) **Patient metadata.** Reintroduce age, sex, and BMI into the risk feature vector once a metadata-linked corpus is available.
- 5) **DICOM ingestion.** Add native DICOM support to preserve metadata, windowing information, and full bit depth.
- 6) **PACS integration.** Wrap inference as a DICOMweb-compatible service that subscribes to study completions on the hospital PACS.
- 7) **Mobile deployment.** Quantise ResNet50 to INT8 and export to TorchScript for on-device inference.
- 8) **Federated training.** Train across institutions without centralising patient data, leveraging differential privacy on aggregate updates.

Multi-body-part and multi-class fracture classification is an orthogonal extension that benefits from all of the above.

XV. CONCLUSION

DeepFracture demonstrates that a compact, explainable, and risk-aware pipeline can be assembled from off-the-shelf components and still produce clinically meaningful behaviour. On the full MURA validation split, ResNet50 with two-phase transfer learning achieves 80.5% accuracy, 0.886 specificity, and 0.852 precision for fracture detection, a substantial gain over a from-scratch baseline and an operating point that supports use as a radiology second reader. The Grad-CAM overlays consistently concentrate on cortical disruptions, providing the visual justification that clinical adoption requires. The radiomic risk module reframes the output from a bare label into an actionable low/medium/high band, and the Streamlit dashboard exposes every intermediate artefact—verdict, confidence, biomarkers, heatmap, error gallery, performance matrices—on a single screen. The error-analysis section is

candid about the cases where the system underperforms, and the future-work agenda is shaped by those failures rather than by abstract aspiration. Taken together, the framework is a useful step towards fracture-detection software that is not only accurate but accountable.

REFERENCES

- [1] P. Rajpurkar *et al.*, “MURA: Large dataset for abnormality detection in musculoskeletal radiographs,” in *Proc. Medical Imaging with Deep Learning (MIDL)*, 2018.
- [2] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [3] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, “Densely connected convolutional networks,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4700–4708.
- [4] M. Tan and Q. V. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 6105–6114.
- [5] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 618–626.
- [6] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [7] R. M. Haralick, K. Shanmugam, and I. Dinstein, “Textural features for image classification,” *IEEE Trans. Systems, Man, and Cybernetics*, vol. SMC-3, no. 6, pp. 610–621, 1973.
- [8] Y. LeCun, Y. Bengio, and G. Hinton, “Deep learning,” *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [9] S.-H. Kong *et al.*, “A novel fracture prediction model using machine learning in a community-based cohort,” *JBM R Plus*, vol. 4, no. 3, e10337, 2020.
- [10] A. Krizhevsky, I. Sutskever, and G. E. Hinton, “ImageNet classification with deep convolutional neural networks,” in *Adv. Neural Information Processing Systems (NeurIPS)*, 2012, pp. 1097–1105.
- [11] K. Simonyan and A. Zisserman, “Very deep convolutional networks for large-scale image recognition,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2015.
- [12] C. Szegedy *et al.*, “Going deeper with convolutions,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 1–9.
- [13] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *Proc. MICCAI*, 2015, pp. 234–241.
- [14] G. Litjens *et al.*, “A survey on deep learning in medical image analysis,” *Medical Image Analysis*, vol. 42, pp. 60–88, 2017.
- [15] A. Esteva *et al.*, “Dermatologist-level classification of skin cancer with deep neural networks,” *Nature*, vol. 542, no. 7639, pp. 115–118, 2017.
- [16] B. Zhou, A. Khosla, A. Lapiedriza, A. Oliva, and A. Torralba, “Learning deep features for discriminative localization,” in *Proc. IEEE CVPR*, 2016, pp. 2921–2929.
- [17] S. M. Pizer *et al.*, “Adaptive histogram equalization and its variations,” *Computer Vision, Graphics, and Image Processing*, vol. 39, no. 3, pp. 355–368, 1987.
- [18] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *Proc. ICLR*, 2015.
- [19] A. Paszke *et al.*, “PyTorch: An imperative style, high-performance deep learning library,” in *Adv. NeurIPS*, 2019, pp. 8024–8035.
- [20] F. Pedregosa *et al.*, “Scikit-learn: Machine learning in Python,” *J. Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [21] A. Dosovitskiy *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *Proc. ICLR*, 2021.
- [22] Streamlit Inc., “Streamlit: The fastest way to build data apps in Python.” [Online]. Available: <https://streamlit.io>